

Jawless fishes

PHYLUM	Chordata
CLASSES	Cephalaspidomorphi
ORDERS	2
FAMILIES	4
SPECIES	About 121

The earliest fish appear in the fossil record about 530 million years ago. They were all jawless and had a very different appearance to the fishes we know today. Four hundred million years ago, jawless fishes were abundant and diverse, but today there are only two small living groups: hagfish and lampreys. These are superficially similar, with an elongated body, smooth, slimy skin, and of course, no proper jaws, but the exact relationship to each other is still under debate. Hagfishes are found only in the ocean, in deep temperate and cold waters. Lampreys are found both in cool coastal water and in fresh water, mostly in the northern hemisphere.

Anatomy

Jawless fish do not have biting jaws, something that every other vertebrate animal does. Instead, lampreys have a round, suckerlike mouth armed with concentric rows of rasping teeth made of keratin, while hagfish have a slitlike mouth with teeth on their tongue. The simple skeleton of a jawless fish is made entirely of flexible cartilage, including a soft cranium instead of a hard skull, and a simple tubelike notochord instead of articulating vertebrae. Lampreys may have a few rudimentary vertebrae but hagfish do not and so cannot be called vertebrates at all. This has led to hagfish being grouped with invertebrates in the past. The gills of hagfish open through 1 to 16 circular pores,

while lampreys have 7. While lampreys have moderately good vision, and use it to home in on their victims, hagfish have only vestigial eyes but an excellent sense of smell for detecting carrion and small invertebrates.

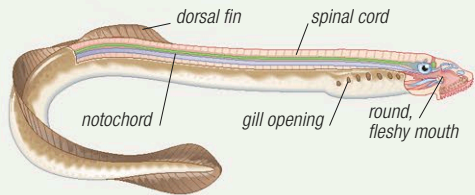
Reproduction

Most lampreys live and breed in fresh water but marine species only enter fresh water when it is time to spawn. They swim up rivers to clean gravelly areas, where spawning pairs excavate shallow nests, lay their eggs and then die. The eggs hatch into ammocoetes larvae that drift downriver to muddy areas where they excavate and live in burrows. They feed by filtering microscopic organisms from the water and by sucking in detritus. After 6-8 years they gradually change into

the adult form and migrate out to sea. In contrast, hagfish have no larval stage and the young hatch from eggs laid on the sea bed.

BODY SECTION

The bodies of lampreys are supported by a notochord that lies below the spinal cord. They also have rudimentary vertebrae made of cartilage.



PARASITISM

Adult marine lampreys use their suckerlike mouthparts to attach themselves to other fish. Once firmly attached, they rasp off flesh and blood, at times enough to kill their host. Most freshwater species are nonparasitic and eat small invertebrates.



Myxine glutinosa

Hagfish



Length Up to 16 in (40 cm)
Weight Up to 27 oz (750 g)
Sex Male/Female
Status Least concern

Location North Atlantic, Mediterranean



Despite having a simple cartilaginous skull, hagfish lack an intact vertebral column. They have been classified with the lampreys (see p.472) as "jawless fish," but are now considered not to be closely related. The bodies of hagfish ooze defensive slime, especially when captured. They can knot their body to free the slime and to assist feeding. A knot formed near the tail works its way up to the head, forcing the mouth away with a lump of its prey's flesh.



ATLANTIC HAGFISH

This species of hagfish lives in the North Atlantic and Mediterranean where it buries itself in mud and feeds on dead and dying fish.

Petromyzon marinus

Sea lamprey



Length Up to 4 ft (1.2 m)
Weight Up to 5½ lb (2.5 kg)
Sex Male/Female
Status Least concern

Location North Atlantic, Mediterranean, North America, Europe



The adult sea lamprey is a parasite that attacks a wide range of saltwater fish, including salmon, trout, herring, mackerel, and some sharks. Clinging to prey by its sucker teeth, it rasps flesh and gains blood using its rough-surfaced tongue. It migrates from the sea to breed in freshwater, but is landlocked in the North American Great Lakes. The opening of the Welland Ship Canal in 1929 allowed lampreys throughout the Great Lakes, and they had a devastating effect on populations of lake trout and other fishes. Since then, the sea lamprey population has been controlled.

SUCKERS



Sea lampreys have oral discs called "suckers" in place of their mouths. The sucker, which is wider than the fishes's body, has a fringed edge and contains many small teeth arranged in concentric rows, with larger teeth surrounding the mouth opening.

EEL-LIKE BODY

Both the adult and the larvae of the sea lamprey have eel-like bodies, which are rounded near the head but flattened at the tail.

